

Aquarry

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Aquarry, Inc.

Company or organization location (we welcome applicants from anywhere in the world)

Golden, CO

Name(s) of primary point(s) of contact for this application

Kate Murphy

Brief company or organization description <20 words

Aquarry converts flooded surface mines—"pit lakes"—into massive, permanent carbon sinks.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Aquarry works to convert pit lakes, the water bodies which form in pit mines after closure, into carbon sinks. Our technology relies on geochemical techniques to draw CO₂ from the atmosphere into the water, where it eventually forms minerals which sink to the lakebed. In doing so, it removes some

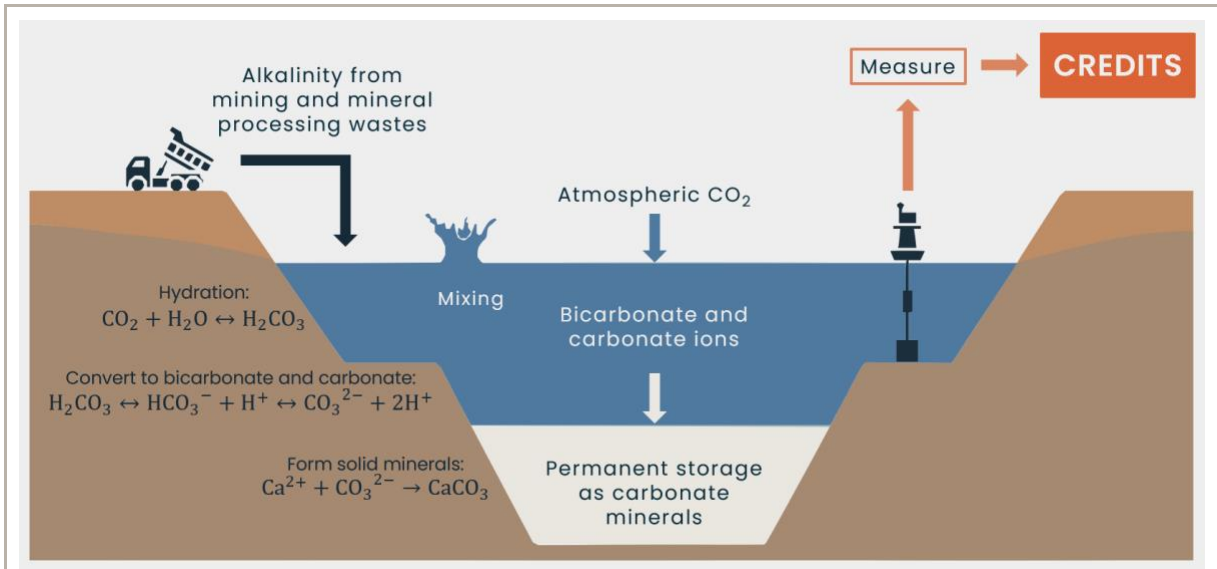
¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

dissolved metals and backfills the pit with material partially drawn from the air, aiding in achieving mine closure goals. CO₂ is stored permanently, is easily measured and monitored, and uses little energy or valuable land. Since pit lakes can be extremely large, with some well over 100 billion liters of water, the technology enables high cumulative storage per site: for larger mines, cumulative storage capacity is >10 MtCO₂. This in turn keeps costs low, <\$80/tCO₂ when site conditions are optimal. Globally, there are many thousands of pit lakes, and we expect total available capacity >50 GtCO₂.

Our technology leverages some of the core principles underlying OAE, ERW, and mineralization and applies them in post-mining pit lakes. Pit lakes form after the closure of open pit mines, which can be kilometers wide and hundreds of meters deep. Many open pit mines intersect the groundwater table, and during operation the mine must be continuously pumped dry. When the mine is closed, the pumps are turned off and the water level in the pit eventually reaches equilibrium with the groundwater level, resulting in the creation of a pit lake. Pit lake water quality depends on the chemistry of the rock walls and groundwater and can span a wide range: some lakes have near-drinking water quality whereas others can be highly acidic with high concentrations of dissolved metals. Additionally, most lakes feature physical hazards due to steep, unstable side walls. While some pit lakes may be remediated, recontoured, and used for recreation or fisheries, few beneficial uses exist.

Aquarry's technology resembles OAE in that we dissolve Ca- or Mg-rich minerals in water, increasing pH and converting aqueous CO₂ to bicarbonate. This depletes the water of CO₂ with respect to the atmosphere. CO₂ is then drawn down into the water according to Henry's law until equilibrium with the air is re-established. *Unlike* OAE, we deliberately drive the concentrations of dissolved carbon and alkaline metals to saturation, at which point carbonate minerals such as calcite (CaCO₃) form and sink to the bottom of the lake where they remain indefinitely. Precipitation of carbonates may cause net emission of CO₂ in marine applications, where the ocean is already supersaturated in calcite and carbon is stored in solution. Our technology does not face these constraints. Potential alkalinity loss associated with precipitation of endogenous dissolved Ca²⁺ or Mg²⁺ in a pit lake is a minor penalty compared to subsequent capture and storage in mineral form over a longer period of time.

Implementation: Aquarry's process is depicted in the diagram below. First, potential sites are analyzed and modeled to determine suitability. Factors determining suitability include hydrology, water and ore body chemistry, water volume, proximity of alkalinity, and pre-existing infrastructure such as a grid connection and roads. We then source alkaline materials available on or near the mine site. These materials may comprise alkaline wastes such as mine tailings, steel slag, or cement kiln dust. Selection of feedstocks is based on reactivity, cost, location, and mineralization capacity. Materials are transported to the site (if not already present) and incrementally dissolved with pit lake water to increase pH. The alkaline stream is discharged to the lake via a passive aeration system to speed absorption of CO₂. As concentrations of Ca²⁺ or Mg²⁺ and CO₃²⁻ increase, carbonate minerals will passively form and settle to the lakebed. This represents a permanent means of storage as a carbonate sediment.



The process is monitored via in situ sensors which track parameters such as pH, pCO₂ and specific conductivity. These real-time data are used to inform our models and process before adding the next dose of alkalinity. Additionally, we periodically collect samples of water and sediments for laboratory analysis. Results of these analyses are used to quantify carbon removed, optimize the process, and in reports to regulators.

Differentiation: Aquarry is the first group to propose the use of pit lake alkalization for CDR—no academics, companies, or other organizations are in the space. No other companies are using pit lakes for CDR by any means. Further, PLAE is distinct from other CDR technologies in blending the best aspects of OAE and ERW. Like OAE, we alkalize water bodies to induce CO₂ absorption from the atmosphere, but we do so in a manner that is far less environmentally risky and far more easily measured since we do not suffer from the extreme dilution which occurs in the ocean. As in ERW, we source alkalinity from silicate minerals or industrial wastes and store CO₂ as durable carbonate minerals. However, as opposed to ex situ ERW we require no reactors, high temperatures, or concentrated CO₂, keeping capex and energy costs low. We also avoid the downsides of surficial ERW, which requires large areas of land and faces measurement challenges due to runoff.

The ideal CDR technology can store gigatons of carbon for thousands of years, costs <\$100/tCO₂, uses little energy, is measurable, has co-benefits, and minimizes use of high-value land. All existing technologies have tradeoffs, e.g., reforestation is low cost and has co-benefits, but is not durable or easy to measure. Ours is the only solution which meets or exceeds all these criteria: we can capture and permanently store gigatons of CO₂ for <\$80/t, with facile measurement, low energy requirements, while utilizing degraded land and cleaning up toxic mine sites. While the planet needs an “all of the above” approach to stabilize the climate, our technology is unique in its potential to provide CDR of the highest value at a low cost.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier’s cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

² We’re looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don’t quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

Aquarry is building the world's first demonstration of carbon removal via alkalization of a mine pit lake. For this proof of concept project we are working with Winsome Resources to build the first CDR hub on a mine site. Winsome is acquiring the inactive Renard diamond mine for the infrastructure installed at the site, which they intend to use to process lithium ore from a future mine ~40 km away. Since the site has >20 Mt kimberlite tailings they are working with CDR startups to utilize them for CDR. Aquarry is the only organization working with the current pit lake, R65, which may be seen in the diagram below.



Costs: Projected costs for this project are <\$500/tCO₂ and <\$80/tCO₂ at scale. Costs for this project are dominated by capex (65%) and electricity (25%). The primary driver of cost reduction is in moving into larger lakes, where capacity is far higher and we achieve economies of scale with regards to capex and labor. The pit lake at Renard is relatively small, with CO₂ capacity <100kt, whereas far larger pit lakes with capacity >10Mt exist. We also expect electricity costs to be lower in subsequent projects where there is a grid connection or renewables, as opposed to Renard's dedicated LNG plant.

Scale: For this project we are working with a lake with capacity of 64ktCO₂. Assuming a project lifetime of 5 years, we will remove 12 ktCO₂/yr (gross). For the much larger lakes we plan to work with in the future, we assume an average volume 40M m³, with average capacity 7.5 MtCO₂ per lake. If each lake has a 15 year lifetime (i.e., we capture on average 500 ktCO₂/year/lake), we can achieve 100 Mt/yr by working in just 200 pit lakes. This is a reasonable number considering there are tens of thousands of lakes globally (of all sizes).

Quantification: Carbon removed is measured by determining carbon content and thickness of sediments via periodic sampling and laboratory analysis via combustion and/or acid digestion. We additionally measure dissolved carbon in the water via in situ pH and pCO₂ sensors, with periodic validation via laboratory measurements of water samples. These measurements are direct and not heavily reliant on models except in interpolating between discrete measurement locations.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

The biggest risks we've identified are:

1. Variable CO₂ transfer rate between the atmosphere and the lake water. CO₂ is driven into water according to Henry's law due to disequilibrium between atmospheric and dissolved (aqueous) CO₂. The transfer rate depends on concentration difference, surface area, and environmental factors. Through excess alkalinity addition we can achieve high atmosphere-solution concentration gradients, which will speed transfer. We also speed transfer by artificially increasing surface area using fountains, sprayers, or falling film aerators. However, we have no control over rain, wind, and temperature, which will substantially affect the transfer rate. To mitigate this risk, we design for the "worst case" scenario in which rain and wind are historically low.
2. Environmental hazards. While pit lakes are relatively closed systems, they may have some amount of discharge to groundwater or surface waters. Potential sources of contaminants include our alkaline feedstock or the host orebody itself, if particular species become mobilized during treatment. In general, we can mitigate these risks by careful selection of the site and alkaline materials combined with thorough sampling and modeling before commencing treatment. Once water treatment has begun we will regularly sample water from multiple locations within the lake and analyze for contaminants of concern. At Renard, feedstocks are relatively clean and all water outflows from the pit lake pass through a water treatment plant before discharge to the environment, providing an additional layer of safety.
3. The diversity and complexity of water quality regulations, which vary between states, provinces, and countries. Navigating the numerous regulatory agencies as we expand operations may present a significant challenge. Furthermore, while we expect to maintain or improve existing water quality, our process represents a departure from standard mine water treatment and beneficial uses of pit lakes are generally not recognized. To mitigate these risks, we will engage with consulting firms which work on mine remediation across different geographies.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	40,550
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Q3 2027
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$434
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$500

³ This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).